Technical Note

Power Electronics and Control Systems

Active power decoupling of a single-phase AFE

the ripple port and its control in the model `single_phase_afe`

Why the power of a single-phase converter pulsates at twice the grid frequency, how the model ***`single_phase_afe`*** *of the repository* ***`modelization-and-control`*** *moves that pulsation from the DC link into a buffer capacitor through a second full bridge, and how the control loops that do it are built. With a check of the numbers of the model.*

Authors:	**Revision:**	**Date:**
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Abstract: A single-phase converter at unity power factor takes from the grid a power that pulsates at twice the grid frequency: it goes from zero to twice the average power, one hundred times per second on a 50 Hz grid. On a three-phase converter the three pulsations cancel; on a single-phase one they reach the DC link, where they become a voltage ripple at 100 Hz. The usual cure is a large DC-link capacitor. The model `single_phase_afe` does something else: a second full bridge, the *ripple port*, connects a buffer capacitor to the DC link through two inductors, and a control loop makes this bridge draw from the DC link exactly the pulsating part of the power, so that the DC link sees only the average. This note explains the pulsation and the energy it moves, the circuit of the ripple port and how its parts are sized, and the control of the model step by step: the observer that isolates the ripple, the filter that builds its quadrature signal, the rotation at twice the grid angle that turns the ripple into two constants, the vector PI that drives them to zero, and the reason why the controller output is applied through the β axis. The numbers of the model are checked with a Python script that also runs an averaged simulation of the DC-link side with the discrete loops of the model: the ripple of the DC-link voltage falls from 16 % to below 1 % and the third harmonic of the grid current from 13 % to 0.5 %. The open points found during the check are listed at the end.

Revision history

Rev.	Date	Description	Authors
00	October 2026	First issue: the pulsation, the ripple port, the control chain of the model, check of the numbers, open points.	Davide Bagnara

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Nomenclature

Symbol	Meaning
AFE	active front end: a PWM converter between a DC link and the AC grid, power in both directions
APD	active power decoupling: removing the power pulsation from the DC link with a controlled circuit instead of a large capacitor
ripple port	the second full bridge of the model, with the buffer capacitor on its DC side and its AC side tied to the DC link through two inductors
pu	per unit: a quantity divided by its base value
U, I	peak values of the grid voltage and of the grid current (fundamental)
P, S	average active power and apparent power of the single-phase connection
ω, f	angular frequency $2\pi f$ and frequency of the grid (50 Hz); the pulsation is at 2ω (100 Hz)
θ	grid angle from the PLL; the ripple loop uses 2θ
u_{dc}, $u_{\mathrm{dc,nom}}$	DC-link voltage and its base (1239.4 V); $u_{\mathrm{dc,pu}} = u_{\mathrm{dc}}/u_{\mathrm{dc,nom}}$
C_{dc}	DC-link capacitor (2.4 mF)
u_{b}, C_{b}, L_{b}	buffer voltage, buffer capacitor (3.6 mF) and the two inductors of the ripple port (250 µH each)
i_{b}	current from the DC link into the ripple port (positive when the buffer takes energy)
$i_{\mathrm{b,bez}}$	base of the buffer current acquisition (750 A)
m_{b}	duty (modulation index) of the ripple-port bridge, $u_{\mathrm{dc}} \approx m_{\mathrm{b}}\, u_{\mathrm{b}}$ on average
u_{bez}, i_{bez}	bases of the grid-side quantities of the model (838.4 V, 536.8 A)
t_{s}	control period, 100 µs (5 kHz PWM, double update)
$\alpha\beta$, dq	stationary and rotating two-axis frames; in the ripple loop the frame turns at 2θ
FHT	first harmonic tracker: a resonant observer that extracts the fundamental of a signal and its quadrature
PI, vector PI	proportional-integral controller; the vector PI is a PI on the d and q components with one common output limit
k_{p}, k_{i}	proportional and integral gains, output in pu per pu of error; k_{i} in 1/s

1 Introduction

A converter connected to one phase of the grid has a problem that a three-phase converter does not have: the power it exchanges with the grid is not constant. With a sinusoidal voltage and a sinusoidal current in phase, the instantaneous power is the product of two sinusoids at the same frequency, which is a constant plus a cosine at twice the frequency. The power goes from zero to twice its average value and back, one hundred times per second on a 50 Hz grid. The DC side



of the converter has to take this pulsation. On a three-phase converter the three phases pulsate with 120° between them and the sum is constant; on a single-phase converter there is nothing to cancel it.

The pulsating power has to be stored somewhere. The simplest place is the DC-link capacitor: the energy that comes in during one quarter of the grid period goes out during the next, and the DC-link voltage goes up and down at 100 Hz. To keep the ripple small the capacitor has to be large, and in a 225 kW converter at 1200 V "large" means tens of millifarads of electrolytic capacitors. The alternative is *active power decoupling* (APD): a small extra circuit with its own energy storage takes the pulsating part of the power, and the DC link sees only the average. The storage element can then work with a large voltage swing, which means much less capacitance for the same energy [3, 4, 5].

The model `single_phase_afe` of the repository `modelization-and-control` is a single-phase full-bridge AFE between two phases of a 690 V grid with such a circuit: a second full bridge, called here the *ripple port*, with a buffer capacitor on its DC side and its AC side tied to the DC link through two inductors. A control loop makes the ripple port draw from the DC link a current at 100 Hz that is exactly the pulsating part of the power divided by the DC-link voltage. This note describes how.

Section 2 gives the pulsation and the energy that moves. Section 3 is the circuit of the model and the sizing of its parts. Section 4 is the control, loop by loop, with the reason of each block. Section 5 shows what the decoupling does with an averaged simulation and Section 6 checks the numbers of the model. Section 7 lists the open points. Everything is in simple words; the equations are the few that one needs to size the parts and to set the gains.

2 The power pulsation of a single-phase converter

2.1 The instantaneous power

Take the grid voltage and the grid current as

$$u(t) = U\cos(\omega t), \qquad i(t) = I\cos(\omega t - \varphi), \tag{2.1}$$

with U and I the peak values and φ the phase of the current behind the voltage. The instantaneous power is

$$p(t) = u(t)\,i(t) = \frac{UI}{2}\cos\varphi + \frac{UI}{2}\cos(2\omega t - \varphi) = P + S\cos(2\omega t - \varphi), \tag{2.2}$$

where $P = (UI/2)\cos\varphi$ is the average (active) power and $S = UI/2$ is the apparent power. The second term pulsates at 2ω with amplitude S: at unity power factor it is as large as the active power, and it does not become smaller when the current is reactive. At unity power factor ($\varphi = 0$):

$$p(t) = P\left(1 + \cos 2\omega t\right). \tag{2.3}$$

Figure 1 shows it with the numbers of the model: 225 kW average, 450 kW peak, zero at the zero crossings of voltage and current.

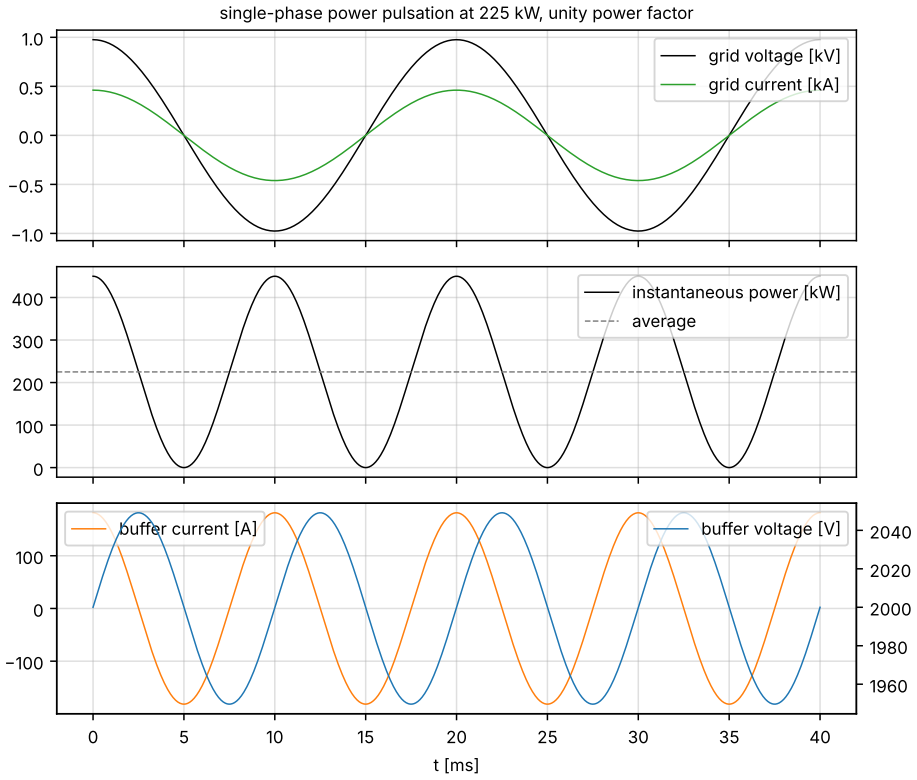


Figure 1: The power pulsation of the model at 225 kW and unity power factor: grid voltage and current (top), instantaneous power (middle), and the current and the voltage of an ideal buffer of 3.6 mF at 2000 V that takes the whole pulsation (bottom). Figure of `single_phase_afe_design_check.py`.

2.2 The energy that moves

The pulsating part of the power, $P\cos 2\omega t$, is the rate at which energy enters and leaves the storage. Its integral is the energy stored:

$$e(t) = \int P\cos(2\omega t)\,dt = \frac{P}{2\omega}\sin(2\omega t), \qquad \Delta E = e_{\max} - e_{\min} = \frac{P}{\omega}. \tag{2.4}$$

This is the number that sizes any storage element, passive or active: at 225 kW and 50 Hz, $\Delta E = 225\,000/314.16 = 716$ J peak-to-peak (875 J at the 275 kW load of the model).

2.3 What the DC link alone would do

If the DC-link capacitor C_{dc} takes the whole pulsation, the current that enters it at 2ω has amplitude P/u_{dc} and the voltage ripple has amplitude

$$\Delta u_{\mathrm{dc}} = \frac{P}{2\omega\, C_{\mathrm{dc}}\, u_{\mathrm{dc}}}. \tag{2.5}$$



With $C_{\mathrm{dc}} = 2.4\,\mathrm{mF}$ and $u_{\mathrm{dc}} = 1239\,\mathrm{V}$ this is 120 V (9.7 %) at 225 kW and 147 V (11.9 %) at 275 kW, with an rms current of 128 A in the capacitor. A ripple of 1 % would need 23 mF. The ripple is not only a stress for the capacitor: Section 5 shows that it also goes through the DC-link voltage controller into the grid current reference and distorts the grid current.

Energy in a capacitor goes with the square of the voltage. A capacitor that swings between $u_{\min}$ and $u_{\max}$ stores

$$\Delta E = \frac{C}{2}\left(u_{\max}^2 - u_{\min}^2\right), \tag{2.6}$$

so for the same ΔE a capacitor that is allowed to swing by ±5 % around 2000 V needs only $716/(0.5 \cdot (2100^2 - 1900^2)) = 1.79\,\mathrm{mF}$. This is why the buffer of an active decoupling circuit can be small: it is not asked to keep its voltage constant.

3 The ripple port of the model

3.1 Circuit

Figure 2 is the power circuit of `single_phase_afe`. From left to right:

- the grid: the three-phase emulator of the library gives the phase voltages of a 690 V, 50 Hz grid; the converter is connected between phases a and b, so it sees the line-to-line voltage u_{ab}, 975.8 V peak;
- the LCL filter: $L_2 = 250\,\mu\mathrm{H}$ on the grid side, $C_{\mathrm{f}} = 200\,\mu\mathrm{F}$ with $50\,\mathrm{m\Omega}$ in series, $L_1 = 250\,\mu\mathrm{H}$ on the converter side; resonance at 1007 Hz;
- the AFE: a full bridge of four IGBTs with antiparallel diodes (Infineon FF1200R17IP5, 1700 V, one half-bridge module per leg), unipolar PWM at 5 kHz;
- the DC link: $C_{\mathrm{dc}} = 2.4\,\mathrm{mF}$, nominal voltage $u_{\mathrm{dc,nom}} = 1239.4\,\mathrm{V}$, and the load, a current source that ramps to 275 kW in 0.25 s;
- the ripple port: a second full bridge, same module, whose two leg midpoints are tied to the two rails of the DC link through $L_{\mathrm{b}} = 250\,\mu\mathrm{H}$ each ($4\,\mathrm{m\Omega}$), and whose DC side is the buffer capacitor $C_{\mathrm{b}} = 3.6\,\mathrm{mF}$ kept at 2000 V.

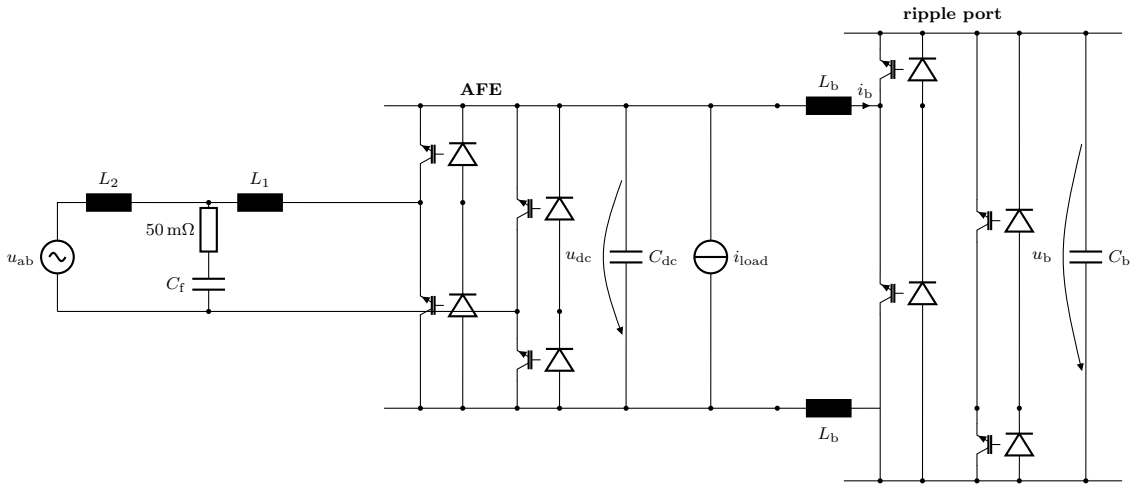


Figure 2: Power circuit of `single_phase_afe`: grid voltage u_{ab}, LCL filter, AFE full bridge, DC link with the current-source load, and the ripple port: a second full bridge whose leg midpoints reach the DC-link rails through the two inductors L_b and whose DC side is the buffer capacitor C_b. Both bridges are drawn with their antiparallel diodes.

3.2 How the ripple port works

Seen from the buffer, the ripple port is a full-bridge DC-DC converter whose "output" is the DC link: with the duty m_b it applies to the inductors a voltage whose average is $m_b u_b$, and the current i_b in the inductors obeys

$$L_{b,tot}\,\frac{di_b}{dt} = u_{dc} - m_b\,u_b - R_b\,i_b, \qquad L_{b,tot} = 2L_b = 500\,\mu H, \tag{3.1}$$

where i_b is counted from the DC link into the port (positive when the buffer takes energy). Note the sign: a larger duty makes the current *smaller*. This is a boost stage seen from the DC link, so the buffer voltage must always stay above u_{dc}: in the model $u_b = 2000\,V$ against $u_{dc} = 1239\,V$, and the steady duty is $m_b = u_{dc}/u_b = 0.62$. The buffer gets the charge $m_b i_b$:

$$C_b\,\frac{du_b}{dt} = m_b\,i_b. \tag{3.2}$$

To decouple the pulsation, the port has to take from the DC link the pulsating part of the power. The current that does it is

$$i_b^*(t) = \frac{P}{u_{dc}}\cos(2\omega t) \quad (182\,A \text{ peak at } 225\,kW,\ 222\,A \text{ at } 275\,kW), \tag{3.3}$$

plus a small DC component that covers the losses of the port and keeps the buffer at its reference. With this current the net current into C_{dc} has no 100 Hz term and u_{dc} is flat (Figure 1, bottom: the buffer takes the energy, its voltage swings from 1950 V to 2049 V, and the DC link does not see it).



3.3 Sizing

Buffer capacitor. From (2.4) and (2.6): $C_\mathrm{b} = 2\Delta E/(u_\mathrm{max}^2 - u_\mathrm{min}^2)$. The model has 3.6 mF (half of the 7.2 mF of the legacy data set) at 2000 V: 716 J give a swing of ±2.5 % (1950 V to 2049 V); the capacitor is oversized by a factor of two for a ±5 % swing (1.79 mF), or it could work at 1500 V with 3.18 mF.

Buffer voltage. It must stay above u_dc at the bottom of the swing, with a margin for the inductor voltage and for transients; 2000 V does that with a wide margin, but it is above the 1700 V class of the module (118 % of the blocking voltage): see Section 7.

Inductors. They carry the 100 Hz current and filter the switching ripple. At 100 Hz, 500 µH drop only 57 V at 182 A, so the current loop needs little voltage. The worst-case switching ripple of a unipolar full bridge at 5 kHz (effective 10 kHz) is $u_\mathrm{b}/(8L_\mathrm{b,tot} \cdot 2f_\mathrm{pwm}) \approx 100$ A peak-to-peak.

Bridge. The current is 182 A peak, 0.24 pu of the acquisition base of 750 A; the module is the same 1200 A one of the AFE, far from its current limit, chosen for the voltage.

Current base. The buffer current is measured with a base of 750 A (`param.afe.buffer.ibez`); the buffer voltage with the DC-link base $u_\mathrm{dc,nom}$, so its reference is $2000/1239.4 = 1.614$ pu. Table 1 collects the numbers.



Table 1: Numbers of the model (`init_model.m`) and derived figures.

Quantity	Value	Note
grid	690 V, 50 Hz	line-to-line a–b, 975.8 V peak = 1.164 pu of u_{bez}
rating / load	225 kW / 275 kW	461 A / 564 A peak; the load is 122 % of the rating
PWM, control	5 kHz, $t_{\mathrm{s}} = 100\,\mu\mathrm{s}$	unipolar, double update; dead time 1 µs
bases	838.4 V, 536.8 A, 1239.4 V	u_{bez}, i_{bez}, $u_{\mathrm{dc,nom}}$ of the legacy single-phase data set
LCL	250 µH, 200 µF, 250 µH	$f_{\mathrm{res}} = 1007\,\mathrm{Hz}$, $f_{\mathrm{s}}/6 = 1667\,\mathrm{Hz}$; C_{f} takes 13 % of P as reactive power
modulation index	0.79	at 225 kW, limit 1
DC link	2.4 mF	ripple alone 9.7 % at 225 kW, 11.9 % at 275 kW
pulsation	182 A, 716 J	100 Hz current amplitude and energy peak-to-peak at 225 kW
buffer	3.6 mF, 2000 V	7200 J stored, swing 1950 V to 2049 V (5 %)
port inductors	2×250 µH, 4 mΩ	57 V at 100 Hz, switching ripple 100 A pk-pk
port duty	0.62	$u_{\mathrm{dc}}/u_{\mathrm{b}}$; the feed-forward of the current loop uses $u_{\mathrm{dc,pu}} \approx 1$ instead
module	FF1200R17IP5	1700 V, 1200 A; $t_{\mathrm{d,off}} + t_{\mathrm{f}} = 1.74\,\mu\mathrm{s}$ at 125 °C

4 The control of the model

4.1 Overview

Figure 3 shows the whole control. Everything runs at $t_{\mathrm{s}} = 100\,\mu\mathrm{s}$, triggered at the bottom and at the top of the 5 kHz carrier. The grid side is a standard grid-following AFE: a PLL gives the grid angle θ, a PI on the DC-link voltage gives the amplitude of the grid current, and a current loop makes the current follow a sinusoid in phase with the grid voltage. The ripple port has three loops: a slow PI on the buffer voltage that sets the DC component of the port current, the *ripple loop* that sets its 100 Hz component, and a fast PI on the port current that applies the sum through the duty of the ripple-port bridge.

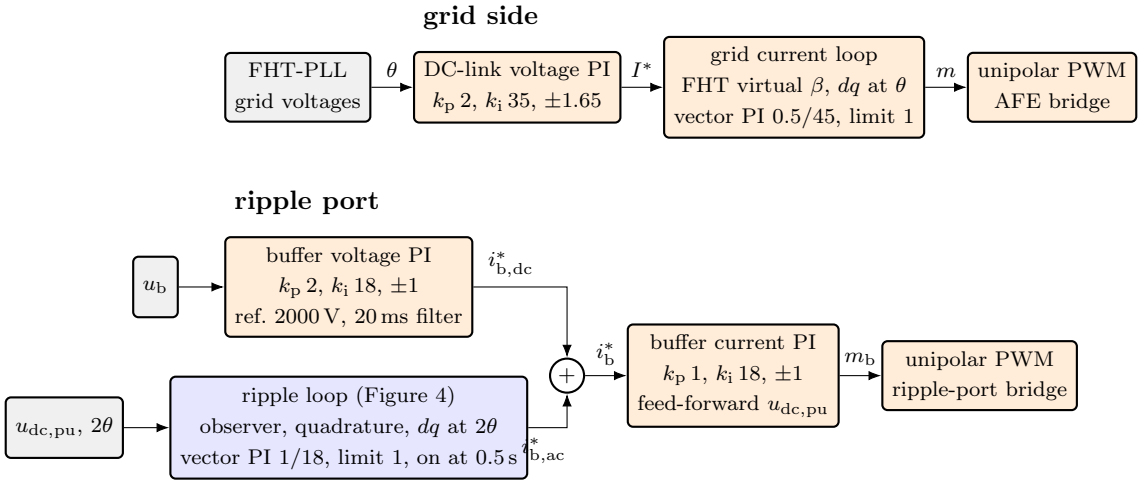


Figure 3: Control of `single_phase_afe`. Grid side: PLL, DC-link voltage loop, grid current loop and PWM of the AFE. Ripple port: the buffer voltage PI gives the DC part of the port current reference, the ripple loop its 100 Hz part, and the buffer current PI applies the sum through the duty of the ripple-port bridge. Gains in pu per pu; k_i in 1/s.

4.2 Grid side

PLL. A first harmonic tracker on each of the three grid voltages, Clarke transform, positive sequence, and a phase observer (`pll_stageB`: PI k_p 1, k_i 80 on the q voltage, double-integrator observer of angle and frequency). The angle θ is the angle of the line-to-line voltage u_{ab} seen by the converter; the ripple loop uses 2θ.

DC-link voltage loop. The reference is 1 pu, applied as a step at 1 ms through a 20 ms low-pass. The PI (k_p 2, k_i 35, output between ±1.65 pu, with anti-windup) gives the amplitude of the grid current I^*, negative when the converter rectifies. The plant is the DC-link capacitor through the power balance: 1 pu of current amplitude changes $u_{dc,pu}$ by 71/s; the crossover is 23 Hz with 79° of phase margin. The PI has no filter at 100 Hz: $|k_p + k_i/(j2\omega)| = 2.0$, so a ripple of 10 % on $u_{dc,pu}$ becomes 0.2 pu of 100 Hz on the current amplitude. This matters when the ripple loop is off (Section 5).

Grid current loop. The measured current is the α component; a first harmonic tracker (a resonant observer at ω with damping $\delta = 0.05$ and observer poles at $-\omega$ and -4ω) gives the fundamental and its quadrature, which is the virtual β. After the rotation at θ the current is a dq vector; the reference is $d = I^*$, $q = 0$ (unity power factor). The vector PI (k_p 0.5, k_i 45, output divided by $u_{dc,pu}$ and limited to 1 as a vector, with the integrators frozen while the limit is active) gives the voltage reference, rotated back to $\alpha\beta$; the α component is the modulation index of the AFE bridge. With the LCL the crossover is 325 Hz with 70° of phase margin; the resonance of the filter at 1007 Hz is below $f_s/6$, the region where the converter-side current feedback damps it by itself, and the 50 mΩ of the capacitor branch help; the gain margin at the resonance is 5 dB.

4.3 The ripple loop, step by step

The ripple loop is the part that does the decoupling (Figure 4). It has to find the 100 Hz ripple of the DC-link voltage, measure its amplitude and its phase, and ask the ripple port for the current that cancels it. The model does it with the same tools used for the grid current: a quadrature pair, a rotation into a frame that turns with the ripple, and a vector PI on the two constants that come out.

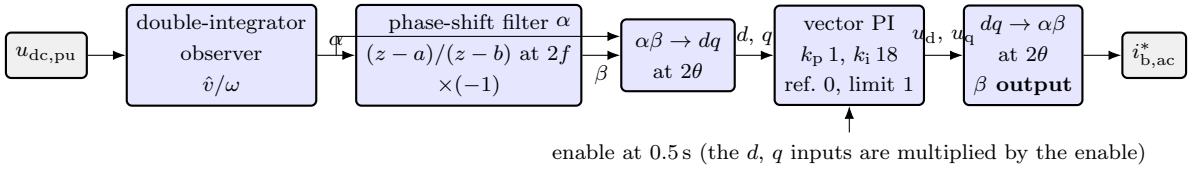


Figure 4: The ripple loop of the model (`dclink_power_decoupling/dclink_decoupling_ctrl`). The observer isolates the ripple of $u_{\mathrm{dc,pu}}$, the phase-shift filter makes its quadrature, the rotation at 2θ turns the pair into two constants, the vector PI drives them to zero and the output is rotated back; the β output is the 100 Hz current reference of the ripple port.

Step 1: isolate the ripple. The measured $u_{\mathrm{dc,pu}}$ is a constant near 1 plus the 100 Hz ripple. The constant must go, or it would be seen by the next blocks as a very large "ripple". The model feeds $u_{\mathrm{dc,pu}}$ to the C function `linear_double_integrator_observer`, an observer of position and velocity of a double integrator driven by the measured signal:

$$\hat{x}_{k+1} = \hat{x}_k + \hat{v}_k t_\mathrm{s} + t_\mathrm{s} k_\mathrm{x}(u_k - \hat{x}_k), \qquad \hat{v}_{k+1} = \hat{v}_k + t_\mathrm{s} k_\mathrm{v}(u_k - \hat{x}_k), \tag{4.1}$$

and takes the velocity output divided by $\omega_{\mathrm{base}} = 2\pi \cdot 50$: $\alpha = \hat{v}/\omega$. The gains come from the observer poles placed at $-\omega$ and -4ω ($k_\mathrm{x} = 1490$, $k_\mathrm{v} = 3.65 \times 10^5$, `afe_ctrl_param`). In continuous time the transfer from u to $\hat{v}$ is

$$H_\mathrm{v}(s) = \frac{k_\mathrm{v}\, s}{s^2 + k_\mathrm{x}\, s + k_\mathrm{v}}, \tag{4.2}$$

a band-pass with natural frequency $\sqrt{k_\mathrm{v}} = 604\,\mathrm{rad/s}$ (96 Hz) and damping 1.23. At DC it is zero, which is what we need; at 100 Hz, which is almost its centre, $H_\mathrm{v}/\omega = 0.78$ with a phase of $-2°$. So the output α is *not* the derivative of $u_{\mathrm{dc,pu}}$ (that would have 90° of phase and a gain of 2) but a copy of the 100 Hz ripple, scaled by 0.78, with its DC value removed and the higher frequencies attenuated. Both readings are fine for the loop: what matters is that α is a clean sinusoid at 2ω proportional to the ripple, and that its gain and phase are known, because they enter the loop gain of Step 5. The observer starts at 50 ms (`time_release`).

Step 2: the quadrature signal. A rotation needs two signals in quadrature, as for the grid current. Here the second one is made by the C function `phase_shift_flt`, the first-order filter

$$y_k = b\, y_{k-1} + \alpha_k - a\, \alpha_{k-1}, \qquad a = 1 + 2\pi t_\mathrm{s} f_\mathrm{c}, \quad b = 1 - 2\pi t_\mathrm{s} f_\mathrm{c}, \quad f_\mathrm{c} = 2f = 100\,\mathrm{Hz}, \tag{4.3}$$



whose frequency response $(z-a)/(z-b)$ has, at f_c, gain 1.03 and phase exactly 90° (127° at 50 Hz, 53° at 200 Hz: it is tuned to one frequency, which is enough because the observer has already removed the rest). The model takes $\beta = -y$. If $\alpha = A\cos(2\theta+\phi)$, then $y = 1.03\,A\cos(2\theta+\phi+90°) = -1.03\,A\sin(2\theta+\phi)$ and $\beta = 1.03\,A\sin(2\theta+\phi)$: the pair (α,β) is a vector of length A that turns *forward* at 2ω, like a positive-sequence pair. The sign change is what makes it turn forward instead of backward. The filter starts at 25 ms.

Step 3: rotation at twice the grid angle. With the pair in hand the model applies the usual Park transform, but with the angle 2θ from the PLL (`alphabeta2dq` with a gain of 2 on the angle):

$$d = \alpha\cos 2\theta + \beta\sin 2\theta = A\cos\phi, \qquad q = -\alpha\sin 2\theta + \beta\cos 2\theta = A\sin\phi. \tag{4.4}$$

The two outputs are constants: the amplitude of the ripple is $\sqrt{d^2+q^2}$ and its phase with respect to 2θ is ϕ. The ripple at 100 Hz is now a DC quantity, and any residual at other frequencies becomes a ripple on d and q that the integrators of the next step average out. The grid angle θ is the right reference because the pulsating power is locked to the grid voltage: at unity power factor the excess power peaks at the peaks of $|u_{ab}|$, that is at $2\theta = 0$.

Step 4: drive the ripple to zero. The vector PI (`dq_vector_pi`, k_p 1, k_i 18, vector output limited to 1 pu) gets the references $d^* = q^* = 0$ and the measured d and q multiplied by the enable (a step at 0.5 s): before the enable it sees no error and its output stays at zero. Its outputs u_d and u_q are the dq components of the current reference of the ripple port, in pu of 750 A. Because the plant is linear, the integrators stop only when $d = q = 0$, that is when the ripple is gone, whatever the exact gain of the chain.

Step 5: back to time domain, through the β output. The output vector is rotated back with the inverse transform at 2θ (`dq2alphabeta`):

$$\alpha_{out} = u_d\cos 2\theta - u_q\sin 2\theta, \qquad \beta_{out} = u_d\sin 2\theta + u_q\cos 2\theta, \tag{4.5}$$

and the model uses β_{out} as the 100 Hz current reference $i^*_{b,ac}$ (the α output goes to a terminator). This is not a detail. In complex notation $\alpha_{out} + j\beta_{out} = (u_d + ju_q)\,e^{j2\theta}$ and $\beta_{out} = \mathrm{Re}\{(u_d + ju_q)\,e^{j(2\theta-90°)}\}$: the current applied to the port is the PI output vector turned by −90°. The DC-link capacitor does the opposite: a current $i_b = I\cos 2\omega t$ taken from the DC link gives $u_{dc} = -(I/2\omega C_{dc})\sin 2\omega t = (I/2\omega C_{dc})\cos(2\omega t + 90°)$, a voltage 90° *ahead*. The two rotations cancel. With the observer (−2°) and the buffer current loop (−7° at 100 Hz) the whole chain, from the PI output back to the detected d, q, is a gain of 0.31 with a phase of −9°: almost a pure gain, which is what a PI wants. Had the α output been used, the chain would have had 81° of phase and the loop would still have converged, but slowly and with a spiral: with a quasi-static complex gain G the closed-loop pole of the envelope is

$$s = -\frac{k_i\,G}{1+k_p\,G}, \qquad G = \underbrace{0.78}_{\text{observer}} \cdot \underbrace{\frac{i_{b,bez}}{2\omega\,C_{dc}\,u_{dc,nom}}}_{\text{capacitor, pu}} \cdot \underbrace{1.005}_{\text{current loop}} \cdot e^{-j\,9°} = 0.31\,e^{-j\,9°}, \tag{4.6}$$



which gives $s = -4.3 + 0.5j$ rad/s: a time constant of 233 ms, with 90 % of the ripple removed about 0.5 s after the enable. With the α output the pole would be $-2.2 - 4.7j$ rad/s: twice as slow and oscillating with a period of 1.3 s.

4.4 Buffer voltage loop

The buffer voltage must stay around its reference while the 100 Hz current goes in and out. A PI (k_p 2, k_i 18, output ±1 pu) on $u_\mathrm{b}/u_\mathrm{dc,nom}$, with the reference 1.614 pu applied at 1 ms through a 20 ms low-pass, gives the DC component $i^*_\mathrm{b,dc}$ of the port current. The plant is the buffer capacitor through (3.2): 1 pu of current changes the buffer voltage (in pu of $u_\mathrm{dc,nom}$) by 104/s; crossover 33 Hz, phase margin 84°. This loop also sees the 100 Hz swing of the buffer voltage (±2.5 %, 0.04 pu of $u_\mathrm{dc,nom}$) and, with $|\mathrm{PI}(j2\omega)| = 2.0$, adds 0.08 pu of 100 Hz to the current reference. It is at the same frequency as the ripple loop output, so the ripple loop absorbs it (its integrators settle where the *total* ripple of u_dc is zero), but a notch at 100 Hz or a moving average over 10 ms on the buffer voltage would take this interaction away.

4.5 Buffer current loop

The sum $i^*_\mathrm{b} = i^*_\mathrm{b,dc} + i^*_\mathrm{b,ac}$ is the reference of a PI (k_p 1, k_i 18, output ±1) on the measured port current in pu of 750 A, whose output plus the feed-forward $u_\mathrm{dc,pu}$ is the duty m_b of the ripple-port bridge. Because of the sign of (3.1) (a larger duty gives a smaller current), the model inverts both the reference and the measurement with a gain of −1 before the PI block, so that the error is $i_\mathrm{b} - i^*_\mathrm{b}$ and the feedback is negative. The plant is the inductor: crossover 848 Hz, phase margin 44°, gain margin 5.9 dB; at 100 Hz the closed loop has gain 1.005 and phase −6.8°, which is the small phase that enters (4.6). The feed-forward is $u_\mathrm{dc,pu} = u_\mathrm{dc}/u_\mathrm{dc,nom} \approx 1$ while the duty of the port is $u_\mathrm{dc}/u_\mathrm{b} \approx 0.62$: the integrator of the PI holds the difference, −0.38, all the time. It is harmless in steady state but it slows the start and it is wrong when u_dc or u_b move; $u_\mathrm{dc}/u_\mathrm{b}$, both measured, is the right feed-forward.

4.6 Timing of the start

In the model the buffer starts at 0 V, the reference of the buffer voltage ramps from zero (step at 1 ms through the 20 ms filter), the observer and the phase-shift filter are held in reset until 50 ms and 25 ms, and the ripple loop is enabled at 0.5 s, when the DC link and the buffer have settled and the load has reached its final value (0.25 s). The whole simulation lasts 1.25 s.

5 What the decoupling does

To see the loops at work without the switching, `single_phase_afe_design_check.py` runs an averaged simulation of the DC-link side: the grid bridge as an average voltage source on the filter inductance (the LCL capacitor is left out), the DC link, the ripple port with its inductors and the buffer, the current-source load ramping to 275 kW in 0.25 s, and the discrete control loops of the model at t_s (DC-link PI, FHT and vector PI of the grid current with the vector clamp, observer, phase-shift filter, rotations, vector PI, buffer PIs with the sign conventions and

the anti-windup of the blocks). The PLL is ideal and the switching ripple is not there. The buffer starts precharged at u_{dc} and the reference filters start from the measured values, so the start is cleaner than in the model. Figures 5 to 8 are the result.

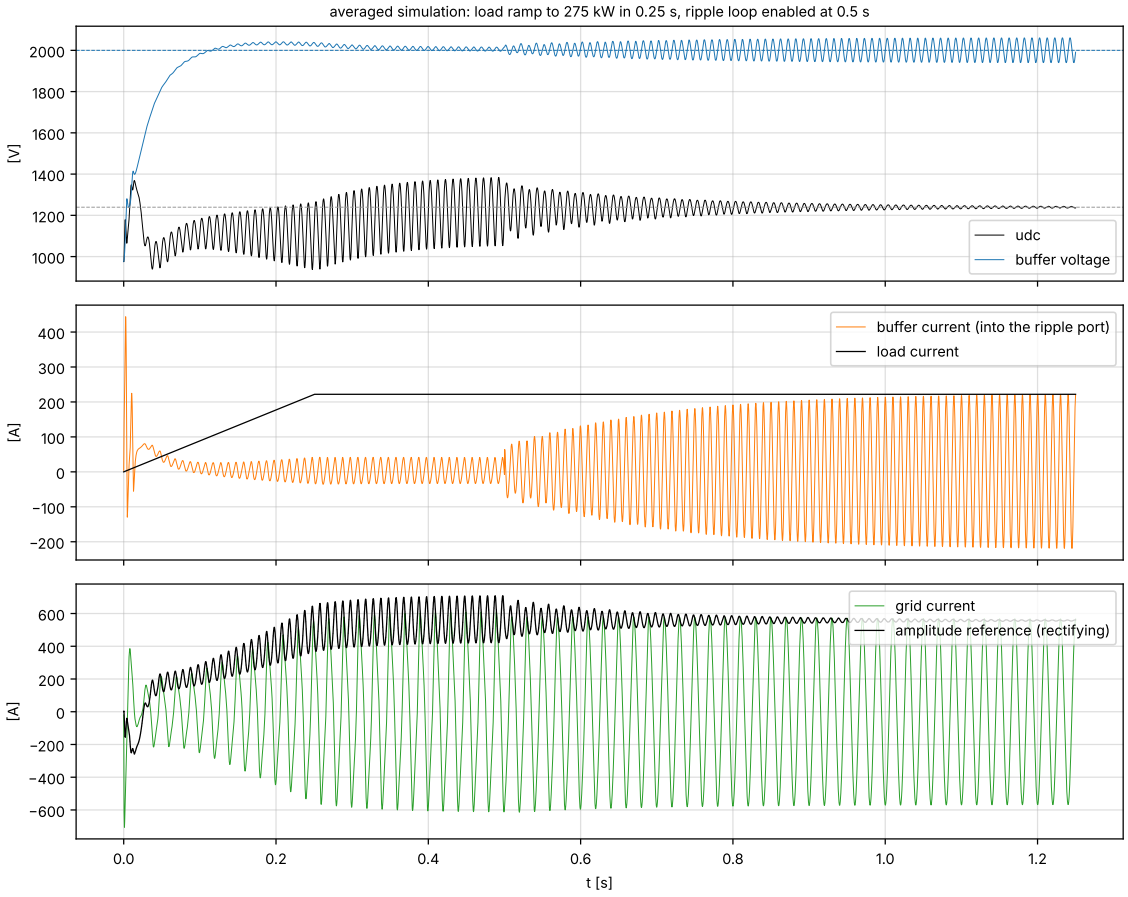


Figure 5: Averaged simulation: DC-link and buffer voltages (top), buffer and load currents (middle), grid current and the amplitude reference of the DC-link PI (bottom). The ripple loop is enabled at 0.5 s.

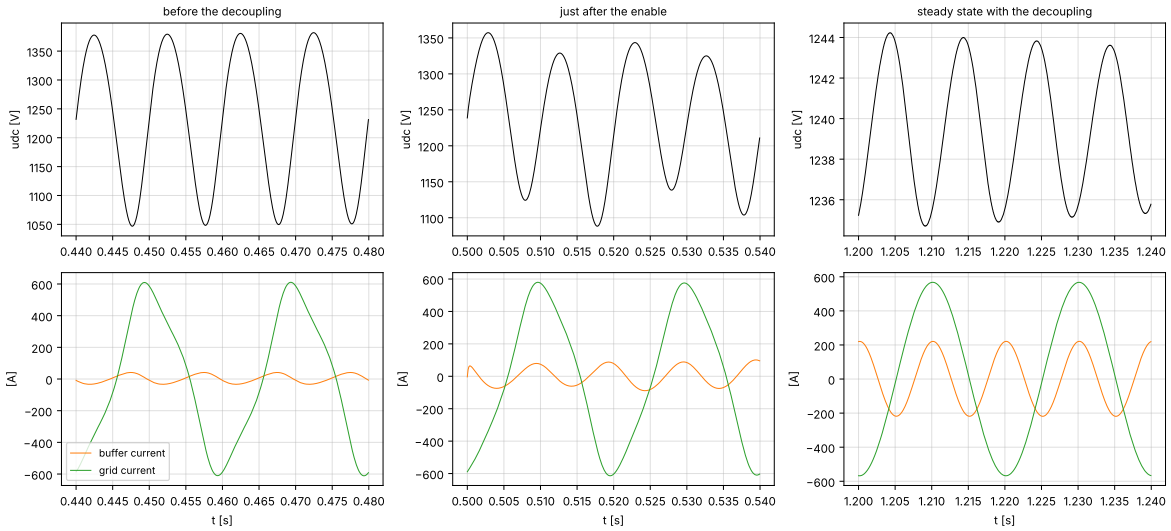


Figure 6: Four grid periods before the enable, just after it and in steady state: u_{dc} (top), buffer and grid currents (bottom).

Before the enable, the DC link ripples by 196 V (16 %, more than the 147 V of the capacitor alone) and the grid current is visibly distorted: the DC-link PI passes the ripple into the current amplitude reference, which oscillates between 400 A and 700 A at 100 Hz (Figure 5, bottom). A sinusoid at ω whose amplitude is modulated at 2ω contains a component at 3ω and one at ω with another phase: the third harmonic is 13.4 % of the fundamental and the THD 13.5 % (Figure 7, left). The extra 50 Hz component also makes the power pulsate more, which is why the ripple is larger than (2.5).

After the enable the port current grows to 221 A peak at 100 Hz, as (3.3) says for 275 kW, in about half a second (Figure 8: the integrators of the vector PI ramp up with the time constant of (4.6); 90 % of the ripple is gone 390 ms after the enable). In steady state the DC-link ripple is 9.5 V (0.8 %), the buffer swings by 121 V peak-to-peak around 2000 V, the duty of the port moves between 0.60 and 0.64, and the grid current is a clean sinusoid: THD 0.5 %. The DC-link voltage itself settles at its reference once the ripple no longer disturbs the voltage PI.

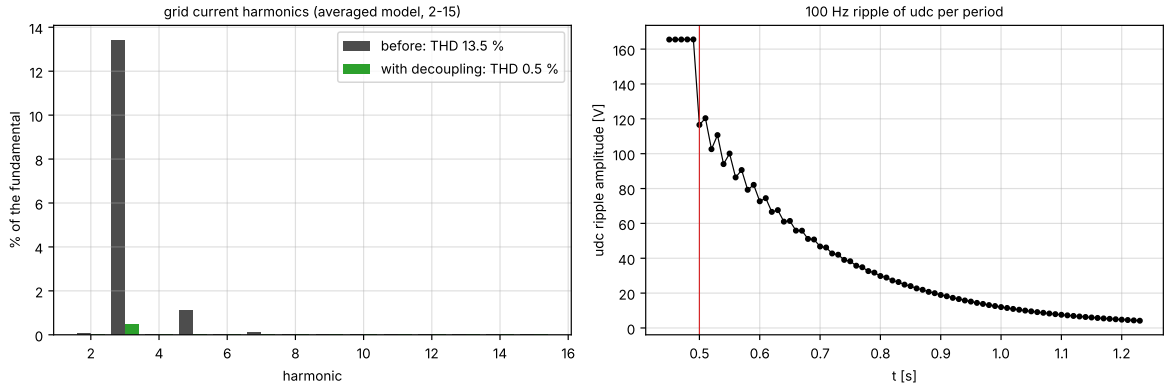


Figure 7: Left: harmonics of the grid current before and with the decoupling (averaged model). Right: amplitude of the 100 Hz ripple of u_{dc}, period by period, around the enable at 0.5 s.

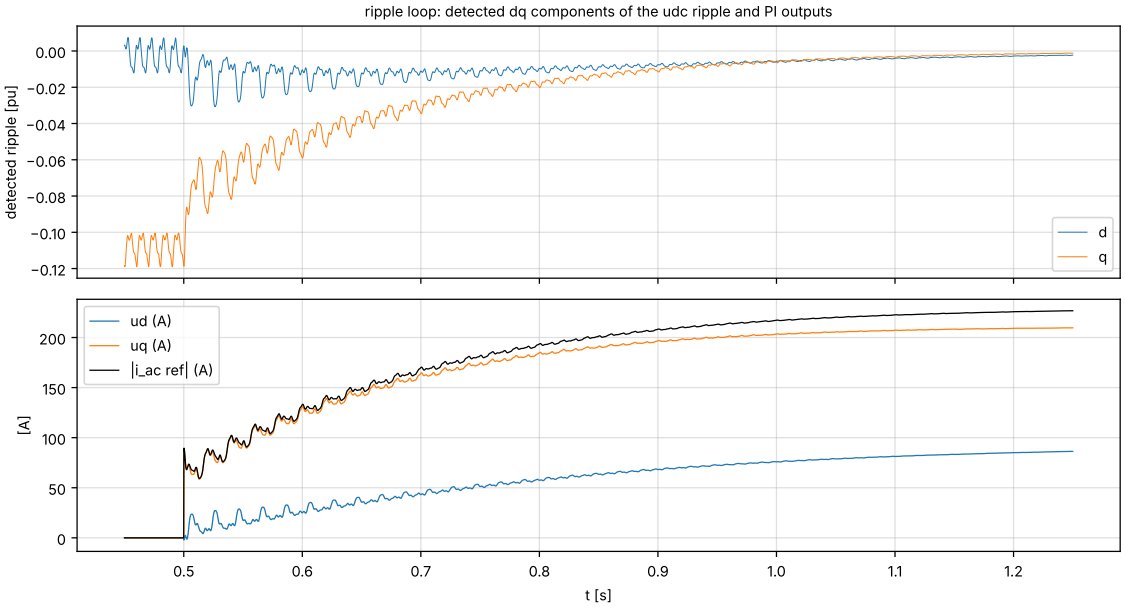


Figure 8: Signals of the ripple loop: the detected d and q components of the ripple (top) and the outputs of the vector PI, in amperes (bottom). The small oscillation on d and q is the residual of other frequencies after the rotation; the integrators average it out.

6 Check of the numbers of the model

The script also computes the frequency responses of the four loops with the values of `init_model.m` (Figure 9, Table 2). The delays of the digital control (one control period of computation plus half a period of PWM) are included.

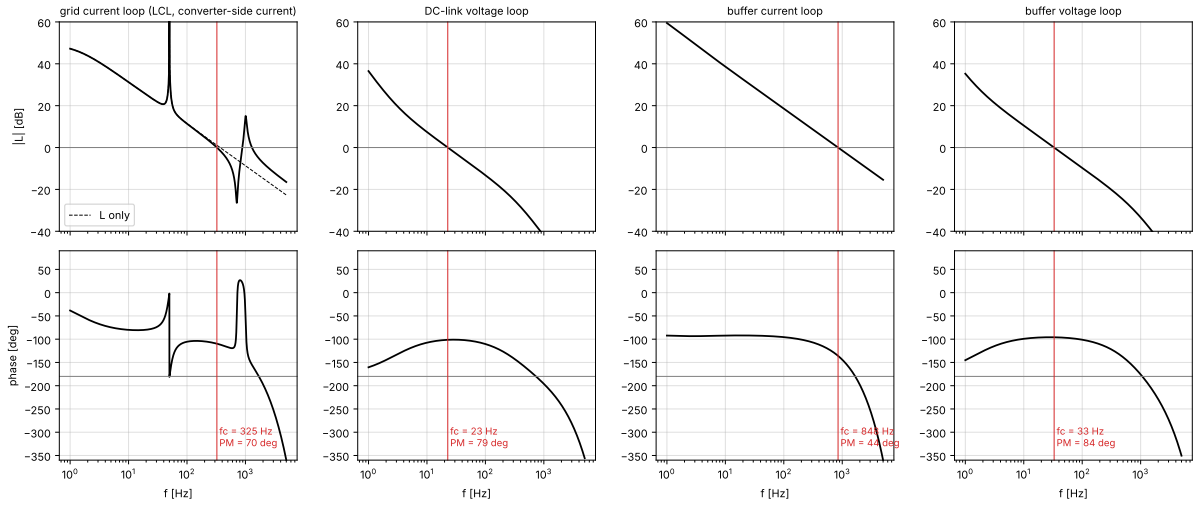


Figure 9: Open-loop frequency responses with the gains of the model: grid current loop with the LCL (and with the inductance alone, dashed), DC-link voltage loop, buffer current loop, buffer voltage loop. Red: crossover and phase margin.



Table 2: Loops of the model: gains and margins.

Loop	Gains	Crossover / margin	Note
grid current	0.5/45, limit 1	325 Hz / 70°	LCL resonance at 1007 Hz with 15 dB of loop gain, gain margin 5.2 dB; PI zero at 14 Hz
DC-link voltage	2/35, limit 1.65	23 Hz / 79°	reference filter 20 ms; $\lvert\mathrm{PI}\rvert = 2$ at 100 Hz
buffer current	1/18, limit 1	848 Hz / 44°	gain margin 5.9 dB; at 100 Hz: 1.005, −6.8°
buffer voltage	2/18, limit 1	33 Hz / 84°	sees the 100 Hz swing of u_b
ripple loop	1/18, limit 1	pole −4.3 rad/s	chain gain 0.31, phase −9°; 90 % in 0.4 s
FHT, PLL	δ 0.05; 1/80	poles $-\omega$, -4ω	observer gains l_1 0.147, l_2 24.3 (FHT); k_x 1490, k_v 3.65×10^5 (ripple observer)

Three things stand out. The decoupling works and the loops are stable with comfortable margins, except the buffer current loop (44°) and the LCL resonance of the grid current loop, both acceptable. The ripple loop is slow: a time constant of 233 ms is fine for a steady load but a load step makes the ripple reappear for about a second; a larger k_i of the ripple vector PI (the chain is almost a pure gain, so k_i sets the pole directly: $k_\mathrm{i} = 100$ would give −24 rad/s) or a feed-forward of the pulsating power computed from the measured grid voltage and current would make it fast. And the DC-link voltage loop has no defence against the 100 Hz ripple: when the ripple loop is off, the grid current carries 13 % of third harmonic.

7 Open points

These are the points found during the check of the model. They are listed here so that they can be decided, not changed silently.

1. **Voltage class of the ripple-port bridge.** The buffer is at 2000 V with a 1700 V module (118 % of its blocking voltage). Options: a lower buffer voltage with a larger swing (1.79 mF give ±5 % at 2000 V; at 1500 V, 3.18 mF), a higher voltage class, or a buck-type port with the buffer below u_dc.
2. **Start-up.** The buffer starts at 0 V: the first milliseconds are a resonant precharge of C_b through the inductors from a DC link already at the rectified grid peak. Both voltage references (DC link and buffer) ramp from zero while the DC link is already at 976 V: the DC-link PI asks for an inverting current until the reference passes the measured voltage. A precharge and references that start from the measured values would fix it.
3. **Feed-forward of the buffer current loop.** $u_\mathrm{dc,pu}$ instead of $u_\mathrm{dc}/u_\mathrm{b}$ (Section 4).



4. **Buffer voltage loop at 100 Hz.** It adds a 100 Hz term to the current reference that the ripple loop has to absorb; a notch or a moving average on u_b would separate the two loops.
5. **Speed of the ripple loop.** Time constant 233 ms; a larger k_i or a power feed-forward (Section 6).
6. **DC-link voltage loop at 100 Hz.** With the ripple loop off the ripple goes into the grid current (13 % of third harmonic); a notch at 100 Hz or a moving average over 10 ms on $u_\mathrm{dc,pu}$ is the usual cure, at the cost of some phase margin.
7. **Dead time.** 1 µs against $t_\mathrm{d,off} + t_\mathrm{f} = 1.74$ µs of the module at 125 °C.
8. **Ratings and bases.** The load is 275 kW on a 225 kW converter; the per-unit bases are those of the legacy single-phase data set (838 V) while the grid peak is 976 V (1.164 pu); the DC-link voltage scaling of the measurement block was 1070 V against $u_\mathrm{dc,nom} = 1239.4$ V (now both read `param.afe.hw.udc_nom`).
9. **PLL and grid.** The PLL reads the three phase voltages of the emulator while the converter is single-phase; a single-phase PLL (SOGI or FHT on u_ab) would match the hardware. The grid has no impedance (no line inductance, no transformer): the LCL resonance and the PLL are not tested against a weak grid.

8 How to run

1. Put the `library` repository on the MATLAB path with subfolders.
2. From the model folder run `init_model.m`: it fills the struct `param` (one group per subsystem, listed in the `README.md` of the folder), opens `single_phase_afe.slx` and checks that every `param.*` name read by the model exists. The parameter functions are in `private/` and are visible only from the model folder.
3. Simulate (1.25 s) and run `plotting_results_sim_results.m`.
4. The numbers and the figures of this note come from the script `single_phase_afe_design_check.py`, run with Python 3 (numpy, scipy, matplotlib) from its folder; the option `--no-sim` skips the averaged simulation.

9 Conclusions

The model `single_phase_afe` implements active power decoupling with a ripple port: a second full bridge, two inductors and a buffer capacitor that swings around 2000 V. The method is a closed loop on the ripple itself: an observer isolates the 100 Hz ripple of the DC-link voltage, a phase-shift filter builds its quadrature, a rotation at twice the grid angle turns it into two constants, a vector PI drives them to zero, and the output is applied through the β axis so that the 90° of the DC-link capacitor are compensated. In the averaged simulation the ripple falls from 16 % to below 1 % and the third harmonic of the grid current from 13 % to 0.5 %, in about half a second. The open points of Section 7 are for the next revision of the model.

10 Files

Table 3: Files of the model and of this note.

File	Content
folder `foundations_of_power_electronics/single_phase_afe/` **of** `modelization-and-control`	
`single_phase_afe.slx`	the model: grid emulator, LCL, the two bridges with the IGBT loss and thermal models, DC link, load, control
`init_model.m`, `private/`	initialization (struct `param`) and its parameter functions; `README.md` describes the folder, the parameter groups and the open points
`claude_analysis/`	`single_phase_afe_design_check.py`, its printed output and the figures of this note
C sources of the library, folder `library/user_defined_functions/ccaller/src/`	
`linear_double_integrator_observer.c`, `phase_shift_flt.c`	the observer of Step 1 and the phase-shift filter of Step 2 (FHT and vector PI: `first_harmonic_tracker.c`, `dqvector_pi.c`; the model uses their Simulink versions)
this note: `repos_doc/single_phase_afe_power_decoupling/` of `repositories-documentation`, copy in `library/documentation/single_phase_afe_power_decoupling/` of `library`	
`single_phase_afe_power_decoupling.tex`, `.pdf`	source and PDF
`single_phase_afe_design_check.py`, `figures/`	the script and the PDF figures it produces



References

[1] pwr-control, *modelization-and-control* repository, https://github.com/pwr-control/modelization-and-control, folder `foundations_of_power_electronics/single_phase_afe`.

[2] pwr-control, *library* repository (Simscape components, C-Caller sources and documentation), https://github.com/pwr-control/library.

[3] P. T. Krein, R. S. Balog, and M. Mirjafari, "Minimum energy and capacitance requirements for single-phase inverters and rectifiers using a ripple port," *IEEE Trans. Power Electron.*, vol. 27, no. 11, pp. 4690–4698, Nov. 2012.

[4] Y. Sun, Y. Liu, M. Su, W. Xiong, and J. Yang, "Review of active power decoupling topologies in single-phase systems," *IEEE Trans. Power Electron.*, vol. 31, no. 7, pp. 4778–4794, July 2016.

[5] H. Hu, S. Harb, N. Kutkut, I. Batarseh, and Z. J. Shen, "A review of power decoupling techniques for microinverters with three different decoupling capacitor locations in PV systems," *IEEE Trans. Power Electron.*, vol. 28, no. 6, pp. 2711–2726, June 2013.

[6] R. Wang, F. Wang, D. Boroyevich, R. Burgos, R. Lai, P. Ning, and K. Rajashekara, "A high power density single-phase PWM rectifier with active ripple energy storage," *IEEE Trans. Power Electron.*, vol. 26, no. 5, pp. 1430–1443, May 2011.

[7] R. Teodorescu, M. Liserre, and P. Rodríguez, *Grid Converters for Photovoltaic and Wind Power Systems*. Wiley-IEEE Press, 2011.

[8] D. Bagnara, *Single-Phase Grid-Tied T-Type Inverter: model, control design and discrete implementation*, pwr-control technical note, Rev. 01, October 2026, `repos_doc/single_phase_grid_tied_inverter_analysis` of https://github.com/pwr-control/repositories-documentation (first harmonic tracker and vector PI on a single-phase current).